

Background and Motivation

- Most data visualization tools provide visual feedback to users.
- Research findings demonstrated that a multimodal display was preferred by users over a unimodal vision-only presentation [1-3].
- Flexible multisensory mapping allows users to mix and match senses (visual, audio-visual, visuo-haptic, audio-haptic, and audio-visual-haptic).
- Virtual Reality (VR) technology can enhance user perception and experience by providing an immersive multimodal environment.
- Multimodal strategies can create a diverse, accessible, and inclusive environment.
- Audience interest and motivation could be substantially enhanced and nurtured if presentations could be combined with a mood of entertainment.

The Proposed Approach

- Users were provided with options to mix and match three senses (i.e., audio, visual, and touch).
- When one modality failed to satisfy user perception and understanding, alternate senses were added to fill the gap.
- If users were overloaded with senses, they could remove sense(s) to find their optimal personalized versions.

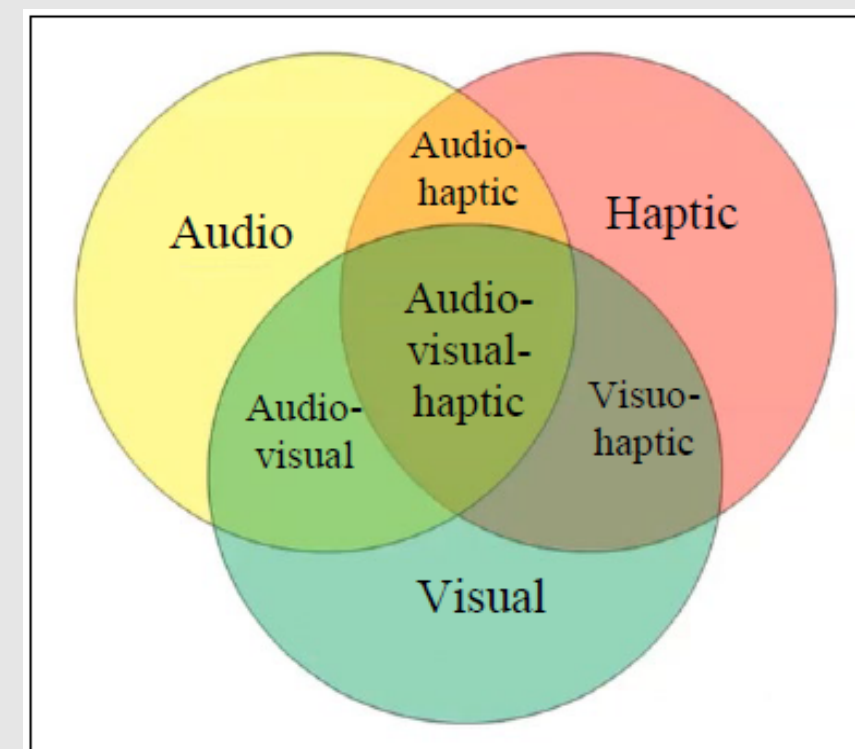


Figure 1. A Venn diagram illustrating different options for combining senses.

- Initially, users used default visual settings, which could be customized with added senses, i.e., audio and touch.
- Users could sonify the application with some popular music or tones.
- Adding touch helped to develop a sense of nearness and a more profound connection in contrast with mouse-based interactions.

Implementation and Results

- The project used Unity3d for its implementation.
- As the program started, an apple with varied colors was displayed. Color variation shows variation in physical properties, i.e., rough or smooth.
- In visuo-haptic mode, users could touch and feel the apple. The reddish part of the model felt rough, the greenish area felt medium-rough (in between rough and smooth), and the bluish part generated a smooth feeling.
- This project used 3d systems' 'Touch' haptic device (www.3dsystems.com) shown in Figure 2.
- A piece of music with varying pitches was played as users continued to explore the model haptically. The pitch of music increased with the values of variables and vice versa.

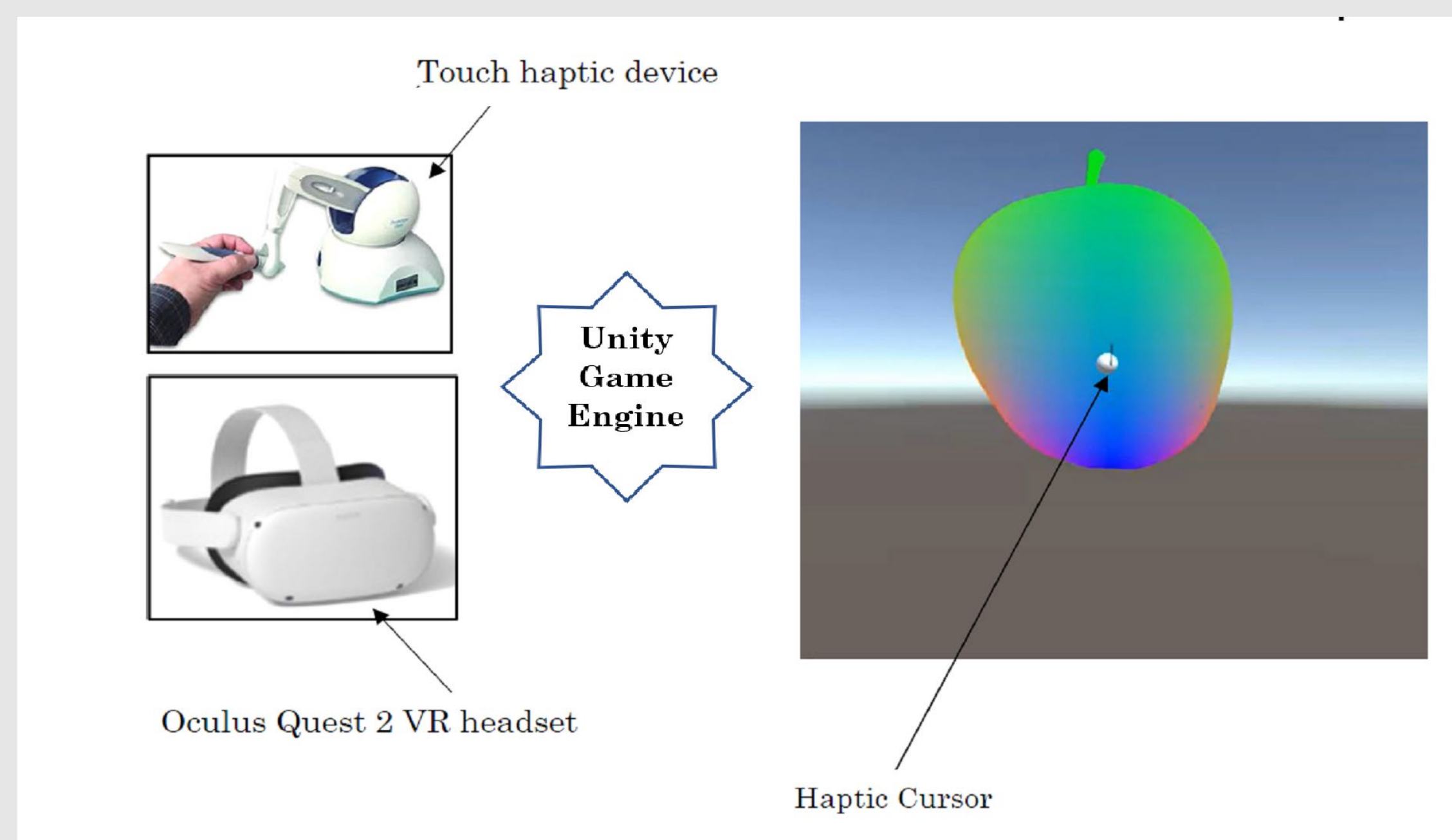


Figure 2. A multimodal display of an apple being explored in an audio-visual-haptic mode. Color variations show variations in surface properties, i.e., friction, which could be perceived when users touched the apple with a 'Touch' haptic device. An Oculus Quest 2 VR headset was used to explore the model in VR. The movement of the haptic device handle is replicated by a haptic cursor in the display represented by a white sphere. As users navigate the apple with a haptic device, they can feel variations in friction and pitch of a music; both friction and pitch increase with values of variables and vice versa.

References

- [1] B. Tomlinson, B. Walker, and E. Moore. Auditory Display in Interactive Science Simulations: Description and Sonification Support Interaction and Enhance Opportunities for Learning. In Proceedings of the 2020 CHI Conference on Human Factors in Computing Systems, pages 1-12 April 2022.
- [2] Robert Bywater and Jonathan Middleton. Melody discrimination and protein fold classification. Heliyon, 2(10):2016.
- [3] S. Panëels and J. Roberts. Review of Designs for Haptic Data Visualization. IEEE Transactions on Haptics, 3(2), pages 119-137, 2010.

Evaluation

- Thirteen sighted subjects (10 males, two females, and one from the LGBTQ+ group) within the age range of 20 to 55 volunteered to participate in the user study.
- The user study focused on participants' levels of immersion and engagement in unimodal versus multimodal environments.
- Participants explored the VR-enhanced and non-VR desktop versions of the application in several modes:
 - vision-only (default),
 - visuo-haptic,
 - audio-haptic, and
 - audio-visual-haptic.
- The unimodal vision-only version (both in VR and non-VR modes) was the least appreciated by participants. Out of 13 participants, 10 preferred the VR-enhanced audio-visual-haptic mode.
- Though participants were divided between the VR-enhanced and non-VR desktop versions, all voted in favor of the multimodal version that included three senses.

Conclusion and Future Work

- In addition to enhanced accessibility, the proposed multimodal presentation incorporated diversity and individualization in the design.
- An in-depth user study and statistical analysis, different objects and their topologies will be explored in a flexible and dynamic multimodal environment.
- Multivariate large data tables will be studied in a multimodal audio-visual-haptic mode.
- The application interface will be customized as necessary so that the BVI and hard-of-hearing (HoH) groups can also be included in the user study and evaluation.